

HAMZA ALTAF

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PROFILE

Computer Science (Cybersecurity) undergraduate at Brunel University London, achieving First Class standing in Year 1. Backend-focused developer with hands-on experience in Java (OOP), JDBC, and SQL, and a working understanding of RESTful architecture, data integrity, and system workflow design. Currently studying Operating Systems & Networks and Software Engineering, building on knowledge of distributed systems, concurrency, process scheduling, and scalable backend design. Comfortable using AI coding tools including Claude and GitHub Copilot to accelerate development and explore new approaches.

TECHNICAL SKILLS

Backend & Languages

- Java (OOP, JDBC), C# (ASP.NET Core), Python, SQL
- RESTful architecture & API design
- CRUD operations & data integrity
- Algorithms & data structures

Systems & Infrastructure

- Relational DB design (JOINS, indexing)
- Operating Systems & Networks, Authentication & access control (JWT, IAM)
- Cloud & endpoint security (MS cert)
- Windows internals & low-level API programming (C)

Frontend & Tooling

- JavaScript, HTML, CSS, React
- Git/GitHub, Jira — Agile sprint & backlog management
- AI coding tools: Claude, GitHub Copilot
- Unit/integration testing
- Technical documentation

Professional Skills

- Agile & Scrum methodology (certified, practised)
- Cross-functional team collaboration
- Analytical debugging & root cause analysis
- Fast adaptation to new languages & frameworks

TECHNICAL EXPERIENCE

Software Developer Intern

Daska City Housing Scheme, Daska, Pakistan • Jun – Jul 2025

- Designed and developed the complete backend of a Real Estate Management Dashboard using Java, JDBC, and SQL — architecting the data layer, implementing end-to-end CRUD operations, and building reliable system workflows for a live production environment used by non-technical staff daily.
- Engineered relational SQL database schemas with attention to normalisation, join performance, and data integrity; wrote multi-table queries with null-safety handling and validation logic.
- Identified and resolved a critical pre-launch data integrity bug: a SQL JOIN was silently returning incorrect results for records with null optional fields; traced root cause through the Java JDBC layer, implemented a null-handling fix, and added regression test coverage.
- Worked across the full development lifecycle — requirements gathering with non-technical stakeholders, system design, iterative Agile development, QA, deployment, and post-launch user training.
- Produced technical documentation including architecture overviews, database schema diagrams, API workflow descriptions, and end-user operation guides.

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

Microsoft Cybersecurity Analyst — Professional Certificate

In Progress

- Cloud security architecture, identity and access management (IAM), endpoint security, vulnerability management, and incident response

Agile Project Management — Google

Completed Sep 2025

- Scrum ceremonies, sprint planning, backlog management, velocity tracking, and iterative delivery in cross-functional teams

EDUCATION

BSc Computer Science (Cybersecurity)

Brunel University London | Final Year: First Class Standing | Expected Jun 2027

- Completed: Introductory Programming (A*), Software Design (A), Data & Information (B+), Logic & Computation (A)
- Currently studying: Databases, Software Engineering, Operating Systems & Networks

Foundation Degree in Science

Kingston University, Kingston upon Thames • Jan – Aug 2024

- Pure Mathematics (82%), Core Physics (84%), Academic English (75%)

PORTFOLIO

SafeCircle — Disaster Support Platform

Oct 2025 – Present | Active | React, JavaScript, HTML/CSS, GitHub, Jira

- Architected multi-role authentication and priority-based alerting logic, routing requests by user role and incident severity.
- Scrum Master: led sprint planning, backlog refinement and retrospectives; maintained the Jira board and Agile documentation.
- Built the Crisis Assistance Page with real-time user-role-based routing, guardian alert workflows, and accessibility features.
- Conducted requirements analysis and user acceptance testing against functional and non-functional criteria.

SwiftBot Shape Detection System

May 2025 | Completed with distinction | Java, SwiftBot API, Algorithms, Hardware Integration

- Built a Java application integrating the SwiftBot hardware API for real-time shape detection, classification, and automated drawing.
- Debugged hardware-software interface issues at the API boundary between the Java application layer and the physical device.
- Applied OOP design principles for a clean, modular, extensible architecture; documented design decisions for peer review.
- Delivered a live technical demonstration to assessors, fielding questions about design decisions.

AI Task Manager API

Jun 2026 | Completed | ASP.NET Core 8, C#, JWT, Repository Pattern

- Built a production-grade REST API in ASP.NET Core 8 with a layered architecture (controllers, services, repositories) and JWT authentication with PBKDF2 password hashing via custom middleware.
- Designed repository and cache abstractions (ITaskRepository, IUserRepository, ICacheService) so the in-memory store and cache-aside layer can be swapped for PostgreSQL/EF Core and Redis
- Implemented full task CRUD with filtering, sorting and pagination, plus an AI-powered task summarisation endpoint behind a stub/live provider abstraction
- Wrote 97 unit and integration tests covering auth, task services, caching, and the full register-login-CRUD-AI-delete lifecycle, all passing.
- Containerised with Docker Compose; configured for one-command deployment to Railway
- GitHub: github.com/HamzaAltaf-799/ai-task-manager-api

Real Estate Management Dashboard

Jun – Jul 2025 | Deployed to production | Java, JDBC, SQL, OOP, Agile

- Full backend system for a live production environment: architected the data layer, implemented CRUD operations, and built end-to-end business logic in Java and SQL.
- Designed a relational database schema with normalisation, complex JOIN queries
- Resolved a critical pre-launch data integrity bug in the JDBC layer with a proper fix and regression coverage.
- Produced full technical documentation and plain-language end-user guides for non-technical operational staff.

Simon Swift — Pattern Recognition Game

Mar 2025 | Completed | Java, Event-Driven Programming, OOP, GitHub

- Designed pattern recognition logic and adaptive difficulty algorithms using event-driven programming in Java.
- Applied OOP principles throughout: clean class hierarchy, separation of concerns, extensible architecture.
- Collaborated in a shared GitHub repository with branch-based workflow, peer code review, and structured commits.
- Produced technical documentation and testing evidence including unit test results and edge case analysis.

win-sandbox-detector

Aug 2026 | Completed | C, Windows API, PEB internals

- Built an educational Windows sandbox and anti-debugging detector in C, implementing 15 checks across four categories (timing, environment fingerprinting, debugger presence, human activity) that feed a cumulative, scored CLEAN/SANDBOX verdict.
- Applied low-level Windows internals including PEB structure fields (BeingDebugged, NtGlobalFlag), NtQueryInformationProcess resolved dynamically via GetProcAddress, and hardware breakpoint detection via thread debug registers.
- Documented the reasoning behind each technique, why layered scored detection beats any single check, and the tool's ethical scope and limitations.
- GitHub: github.com/HamzaAltaf-799/win-sandbox-detector

ADDITIONAL EXPERIENCE

Retail Assistant — John Lewis Partnership (Waitrose)

John Barnes (Finchley Road) • Sep 2025 – Apr 2026 • Kingston • Nov 2024 – Apr 2025

- Delivers consistent, high-quality service in a fast-paced, high-footfall environment, developing communication under pressure, task prioritisation, and stakeholder relationship management skills.
- Regularly handles complex or sensitive customer situations requiring empathy, clear communication, and rapid problem resolution.

INTERESTS & ADDITIONAL INFORMATION

Genuine interest in the technical architecture of online platforms — particularly backend systems for real-time services, account management, and distributed delivery. Follows developments in cloud-native technologies and distributed systems. Active contributor to group coding projects on GitHub. Currently exploring Go fundamentals and distributed systems concepts alongside university studies.